PROJECT DESIGN PHASE-II

Technology Stack, Solution Requirements, Data Flow Diagrams & User Stories

Date	**07 APRIL 2026**
Team ID	69c378a771a975a7e423b048
Project Name	Streamlining Ticket Assignment for Efficient Support Operations
Maximum Marks	**4 Marks**

Objective

This phase focuses on defining the technical foundation, system requirements, and data flow for automating ticket assignment in ServiceNow. It also translates user needs into structured user stories for implementation.

Technology Stack

The solution is built using the following technologies and tools:

- **Platform:** ServiceNow
- **Module:** IT Service Management (ITSM)
- **Automation Tools:**
  - Assignment Rules
  - Data Lookup Rules
  - Flow Designer
- **Scripting (Optional):**
  - JavaScript (for advanced logic in business rules)
- **Database:**
  - ServiceNow Tables (Incident, Task, User, Group)
- **User Interface:**
  - Service Portal / Native UI

Solution Requirements

Functional Requirements

- Automatically assign tickets based on:
  - Category (Network, Hardware, Software)
  - Priority (High, Medium, Low)
  - Location (optional)
- Auto-populate fields using data lookup rules
- Ensure tickets are routed to correct assignment groups
- Trigger notifications after assignment
- Support SLA tracking and compliance

Non-Functional Requirements

- High reliability and accuracy
- Minimal assignment delay
- Scalable for large ticket volumes
- User-friendly interface
- Maintainable and configurable rules

Data Flow Diagram (DFD)

Level 1 DFD

Level 2 DFD (Detailed Flow)

User Stories

Format: *As a [user], I want [goal], so that [benefit]*

IT Support Agent

- As an IT support agent, I want tickets to be automatically assigned so that I can focus on resolving issues instead of routing them.
- As an agent, I want correct categorization so that I receive only relevant tickets.

Service Desk Manager

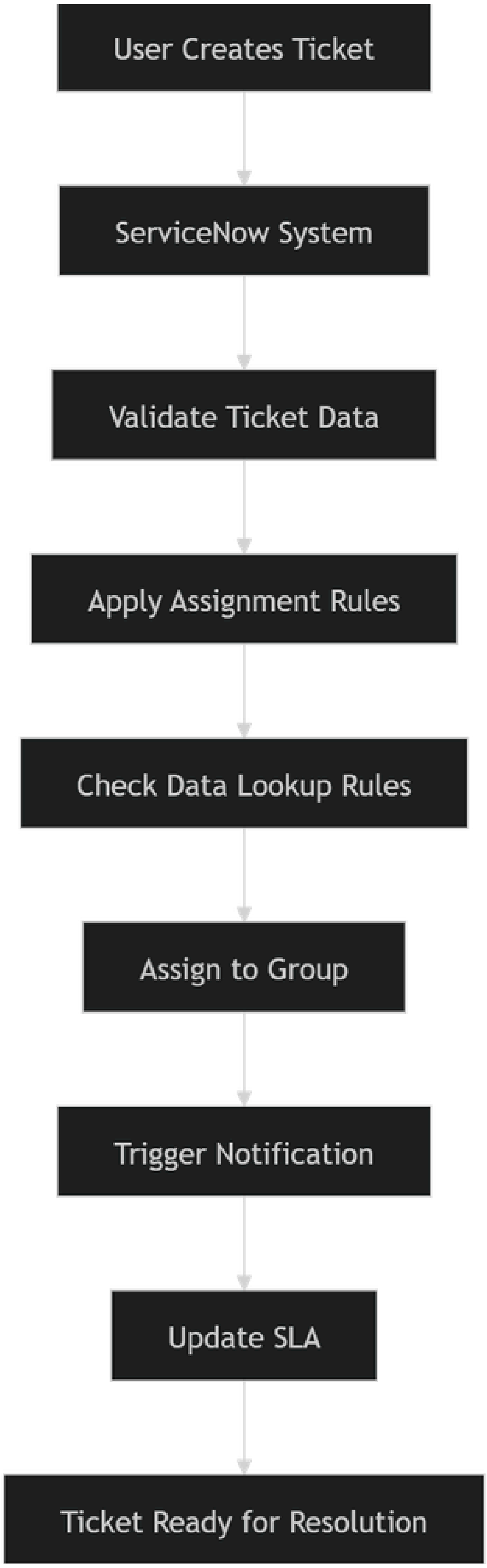
- As a service desk manager, I want automated assignment rules so that team workload is balanced efficiently.
- As a manager, I want SLA tracking so that performance can be monitored and improved.

End User

- As an end user, I want my ticket to reach the right team quickly so that my issue is resolved faster.
- As a user, I want updates/notifications so that I know the status of my request.

System Perspective

- As a system, I should automatically evaluate ticket fields so that assignment decisions are accurate.
- As a system, I should trigger notifications and SLA updates after assignment.



Phase 2 Outcome

- Defined complete technical architecture
- Identified system requirements
- Visualized ticket data flow
- Converted user needs into actionable user stories

In this phase, the focus is on establishing the technical foundation and defining the system requirements for automating ticket assignment within ServiceNow. The solution leverages the IT Service Management (ITSM) module along with built-in automation tools such as Assignment Rules, Data Lookup Rules, and Flow Designer to streamline the ticket routing process. These technologies enable the system to automatically evaluate ticket attributes like category, priority, and location, ensuring accurate and efficient assignment to the appropriate support groups.

The solution requirements are divided into functional and non-functional aspects. Functionally, the system must support automatic ticket assignment, field auto-population, notification triggering, and SLA tracking. Non-functional requirements emphasize reliability, scalability, minimal delays, and ease of maintenance. Together, these requirements ensure that the system not only performs efficiently but also remains adaptable to organizational needs.

To better understand the system behavior, data flow diagrams are used to illustrate how tickets move through the platform—from creation and validation to rule evaluation, assignment, and notification. These diagrams provide a clear visualization of how data is processed and how decisions are made within the system, helping to identify optimization points and ensure logical consistency.

Additionally, user stories are defined to translate business needs into actionable requirements. These stories capture the perspectives of IT support agents, service desk managers, and end users, highlighting their expectations such as faster ticket routing, reduced manual effort, balanced workload distribution, and improved service experience. By aligning technical implementation with user needs, this phase ensures that the proposed solution is both practical and user-centric.

THANK YOU......